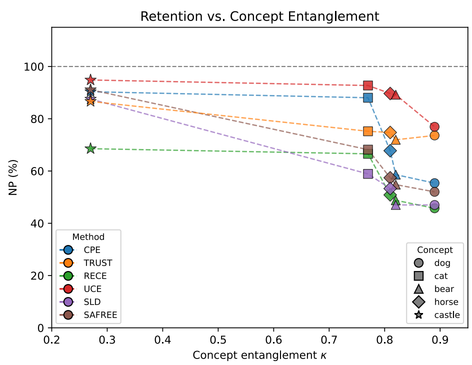
L New Baselines

Table 12: **Additional baseline comparisons.** This table extends the averaged comparison in Table 1 with five newly evaluated baselines: UCE, RECE, CPE, TRUST, SLD, and SAFREE. Methods are grouped by intervention type: closed-form, fine-tuning, or inference-time. Cells report mean $\pm$ standard deviation across the evaluated target concepts, using the same erasure and retention metrics as the main table. The *Original* row denotes the unedited-model reference for this comparison.

		Erasure		Retention			
Method	**Type**	UA↑	IRR↓	IRA↑	NP↑	CP↑	DamageGap↓
uCE	Closed-form	80.6 ± 18.4	10.5 ± 23.4	67.1 ± 12.4	88.6 ± 25.1	99.3 ± 3.1	10.7 ± 25.4
RECE	Closed-form	99.4 ± 0.9	3.0 ± 4.0	43.3 ± 12.1	56.1 ± 10.6	93.5 ± 2.0	37.4 ± 10.2
CPE	FT	99.6 ± 0.9	4.7 ± 9.4	53.4 ± 12.4	71.9 ± 18.1	99.6 ± 2.6	18.7 ± 19.3
TRUST	FT	94.2 ± 2.2	6.5 ± 3.5	53.2 ± 13.7	76.4 ± 9.1	98.0 ± 5.0	11.6 ± 12.2
SLD	Inf.	99.6 ± 0.9	0.5 ± 0.9	35.0 ± 8.1	58.8 ± 16.8	89.1 ± 4.1	30.3 ± 16.3
SAFREE	Inf.	91.4 ± 11.4	8.4 ± 10.2	39.9 ± 8.6	64.7 ± 16.0	85.7 ± 12.2	21.0 ± 19.4



Concept	$\hat{\kappa}$	NP↑	IRA-ratio↑
`castle`	0.27	86.47	0.79
`cat`	0.77	74.88	0.78
`horse`	0.81	62.54	0.74
`bear`	0.82	59.71	0.70
`dog`	0.89	56.58	0.65
Spearman ρ		-1.0	-1.0

Figure 8: **Neighbor preservation scales inversely with $\hat{\kappa}$.** *Left:* NP for each (method, concept) pair. *Right:* New baseline's NP and in-domain retention ratio across the five concepts (Spearman $\rho = -1.0$ for both).

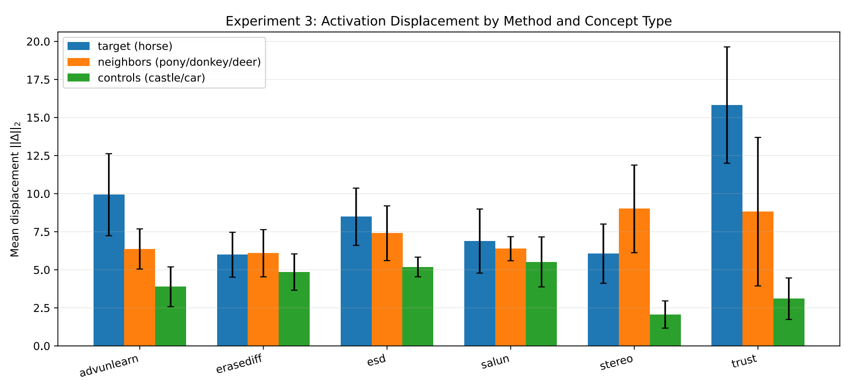


Figure 9: Mean activation displacement $\|\Phi_{\hat{\theta}}(p) - \Phi_{\theta}(p)\|_2$ for six unlearning methods erasing `horse`. Target prompts (blue) confirm Assumption 4.5: erasure requires measurable displacement. Neighbor prompts (orange) are displaced more than control prompts (green) for all methods, consistent with the propagation mechanism in Theorem 4.6.

M New Concepts

Table 13: Results for the five additional target concepts. Original denotes the pretrained model before unlearning. All evaluation metrics are percentages. Higher UA, IRA, NP, and CP are better; lower IRR and DamageGap are better. DamageGap is the difference between control preservation (CP) and neighbor preservation (NP).

Concept	Method	$\hat{\kappa}$	UA (%) ↑	IRR (%) ↓	IRA (%) ↑	NP (%) ↑	CP (%) ↑	DamageGap (%) ↓
Snake	Original	0.37	9.2	6.0	31.1	100.0	100.0	0.0
	EraseDiff	0.37	88.1	8.6	19.1	81.2	93.6	12.4
	ESD	0.37	97.9	6.8	20.6	81.3	98.1	16.8
	SalUn	0.37	100.0	0.0	5.5	72.4	86.2	13.8
	STEREO	0.37	100.0	0.0	14.8	25.1	96.0	71.1
Fish	Original	0.52	6.2	11.0	40.5	100.0	100.0	0.0
	EraseDiff	0.52	92.3	5.3	23.2	78.0	92.1	14.1
	ESD	0.52	94.5	2.7	34.5	79.2	98.0	18.8
	SalUn	0.52	100.0	0.0	14.9	67.5	97.8	30.3
	STEREO	0.52	100.0	0.0	5.8	23.2	38.4	15.2
Tree	Original	0.63	4.4	61.8	51.8	100.0	100.0	0.0
	EraseDiff	0.63	32.0	55.2	53.8	75.1	98.0	22.9
	ESD	0.63	39.7	48.0	50.3	82.4	98.9	16.5
	SalUn	0.63	100.0	0.0	5.8	62.7	88.0	25.3
	STEREO	0.63	100.0	0.0	4.9	26.6	32.0	5.4
Butterfly	Original	0.66	7.1	12.2	36.0	100.0	100.0	0.0
	EraseDiff	0.66	46.2	37.3	34.6	74.7	92.5	17.8
	ESD	0.66	72.6	28.9	31.2	78.1	83.3	5.2
	SalUn	0.66	100.0	0.0	0.9	73.1	93.9	20.8
	STEREO	0.66	100.0	0.0	10.5	19.2	68.0	48.8
Car	Original	0.83	2.6	60.0	87.5	100.0	100.0	0.0
	EraseDiff	0.83	47.4	49.3	69.2	78.7	91.2	12.4
	ESD	0.83	60.5	28.6	75.8	73.8	98.6	24.8
	SalUn	0.83	79.3	16.1	17.8	66.8	85.6	18.8
	STEREO	0.83	99.1	4.3	21.1	16.0	75.2	59.1

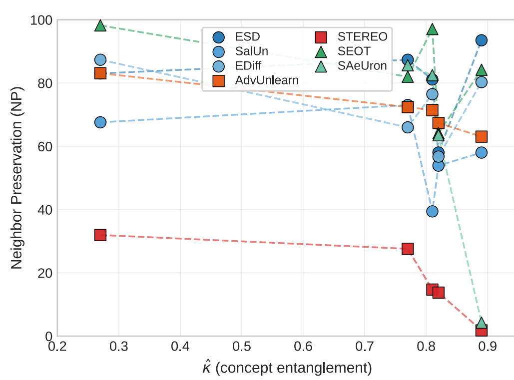


Figure 10: NP for each (method, concept) pair with dashed lines.